



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

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- Methodology
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Executive Summary

Methodologies Used

- **Data Collection:** Utilized API integration and web scraping for comprehensive data gathering.
- **Data Wrangling:** Addressed missing values, standardized variables, and created a new feature (`landing_class`) for easier analysis.
- **Exploratory Data Analysis (EDA):** Investigated launch outcomes by orbit type, payload mass, and booster versions, employing detailed visualizations.
- **Interactive Analytics:** Developed dashboards using Folium and Plotly Dash to visualize outcomes and launch site performance.
- **Predictive Modeling:** Built classification models (Logistic Regression, SVM, Decision Tree, KNN) to forecast landing success.

Key Results

Exploratory Analysis:

- Landing success rates have improved over time.
- Orbits such as ES-L1, GEO, HEO, and SSO demonstrate higher success rates.

Interactive Analytics:

- Kennedy Space Center Launch Complex (KSC LC-39A) has the highest success rate, achieving 76.9% successful launches.
- Payloads between 2,000 and 6,000 kg have the highest success rates, with success diminishing as payload mass increases.

Predictive Analysis:

- Logistic Regression outperformed other models, achieving the highest predictive accuracy for landing success.

Introduction

The emergence of the commercial space age has revolutionized access to space, with companies like Virgin Galactic, Rocket Lab, Blue Origin, and SpaceX making space travel and satellite launches more affordable and accessible. Among these, SpaceX has achieved significant success by offering cost-effective rocket launches, primarily due to its ability to (sometimes) reuse the first stage of its rockets.

Space Y, founded by billionaire industrialist Allon Musk, aims to compete with SpaceX in this innovative industry. To achieve this, the project focuses on addressing key strategic questions:

1. **Launch Pricing:** What should be the competitive price for each rocket launch to establish a foothold in the market?
2. **Landing Success:** What factors influence the likelihood of a successful first-stage landing, a critical factor in cost reduction?
3. **Reusability Prediction:** Under what conditions will SpaceX reuse the first stage of a given launch, providing insights for Space Y's operational strategy?

This project aims to leverage data-driven insights to guide Space Y's decision-making and competitive positioning in the rapidly evolving commercial space industry.

Section 1

Methodology

Methodology

This project utilized a comprehensive approach to gather, process, and analyze data to study factors affecting rocket launch outcomes. The methodology involved the following steps:

- **Data Collection:**
 - Retrieved data using the SpaceX-API and web scraping from the SpaceX Wikipedia page.
 - Consolidated and structured the data for further processing.
- **Data Wrangling:**
 - Handled missing values by applying appropriate replacement techniques.
 - Coded variables consistently and created a new feature, `landing_class`, to categorize landing outcomes.
- **Exploratory Data Analysis (EDA):**
 - Leveraged SQL and visualization tools to examine relationships between launch outcomes and variables such as orbit type, payload mass, and booster version.
 - Used charts to identify trends and patterns in the data.
- **Interactive Visual Analytics:**
 - Developed an interactive map using Folium to analyze launch site performance.
 - Built a Plotly Dash dashboard to visualize success rates by launch site, payload mass, and booster version, enabling deeper insights.
- **Predictive Analysis:**
 - Applied classification models (logistic regression, SVM, decision tree, KNN) to predict landing success.
 - Optimized model performance by finding the best hyperparameters using Grid Search.
 - Evaluated model accuracy on testing data to select the best-performing model.

Data Collection

The data for this report was collected through a combination of API usage and web scraping, ensuring comprehensive and accurate information about SpaceX launches.

SpaceX REST API:

- Data about SpaceX launches was gathered using the SpaceX REST API.
- This API provides detailed information, including the rockets used, payloads delivered, launch specifications, landing details, and outcomes.
- The API delivered data in JSON format, which was then processed and converted into a Pandas DataFrame for analysis.

Web Scraping:

- Python's BeautifulSoup library was used to scrape HTML tables from the SpaceX Falcon 9 launch records available on the SpaceX Wikipedia page.
- The parsed table data was extracted and transformed into a structured Pandas DataFrame for integration with the API data.

Data Collection – SpaceX API



- Use the SpaceX API to get information about the launches using the IDs given for each launch. We will be using columns `rocket`, `payloads`, `launchpad`, and `cores`
 - From the `rocket` column we want to learn the booster name
 - From the `payload` column we want to learn the mass of the payload and the orbit that it is going to
 - From the `launchpad` column we want to know the name of the launch site being used, the longitude, and the latitude.
 - From `cores` we would like to learn the outcome of the landing, the type of the landing, number of flights with that core, whether `gridfins` were used, whether the core is reused, whether `legs` were used, the landing pad used, the `block` of the core which is a number used to separate version of cores, the number of `reused_times` this specific core has been reused, and the `serial` of the core.
- Add the GitHub URL of the completed SpaceX API calls notebook ([must include completed code cell and outcome cell](#)), as an external reference and peer-review purpose

Data Collection - Scraping



- Use the GET command from the requests library to retrieve the HTML page of the SpaceX wiki page
- Use BeautifulSoup to get the table information
- Use some helper functions to parse the table information into similar variables as using the API
- Convert this information into a Pandas DataFrame and save the data to a csv file
- Add the GitHub URL of the completed web scraping notebook, as an external reference and peer-review purpose

Data Wrangling



- True Ocean means the mission outcome was successfully landed to a specific region of the ocean while False Ocean means the mission outcome was unsuccessfully landed to a specific region of the ocean. True RTLS means the mission outcome was successfully landed to a ground pad False RTLS means the mission outcome was unsuccessfully landed to a ground pad. True ASDS means the mission outcome was successfully landed to a drone ship False ASDS means the mission outcome was unsuccessfully landed to a drone ship. None ASDS and None None these represent a failure to land.
- Using this information create a new variable landing_outcome that is 0 if there was a failure and 1 otherwise.
- Add the GitHub URL of your completed data wrangling related notebooks, as an external reference and peer-review purpose

EDA with Data Visualization

Summary of Plots Used and Their Purpose:

- **Bar Charts:**
 - **Why:** To visualize categorical data distributions, such as the frequency of launch outcomes across different orbit types or booster versions. Bar charts are effective for comparing discrete categories.
- **Scatter Plots:**
 - **Why:** To examine relationships between continuous variables, such as payload mass versus success rate or landing outcome. Scatter plots help identify trends or clusters in the data.
- **Line Plots:**
 - **Why:** To track trends over time, such as the success rate of landings by year. Line plots are useful for time-series analysis.
- Add the GitHub URL of your completed EDA with data visualization notebook, as an external reference and peer-review purpose

EDA with SQL

- **Unique Launch Sites**
- **Filter Launch Sites Starting with "CCA"**
- **Total Payload Mass by NASA (CRS)**
- **Average Payload Mass for Specific Booster Version**
- **First Successful Ground Pad Landing**
- **Booster Versions with Successful Drone Ship Landings and Specific Payload**
- **Count of Successful and Failed Mission Outcomes**
- **Boosters with Maximum Payload Mass**
- **Drone Ship Failures in 2015**
- **Rank Landing Outcomes by Count**
- **Add the GitHub URL of your completed EDA with SQL notebook, as an external reference and peer-review purpose**

Build an Interactive Map with Folium

- **Markers Representing Launch Sites:**
 - Markers were created and added to represent each launch site. These markers provide an interactive way to pinpoint the exact geographical location of the launch sites on the map.
- **Circles Showing Proximity Metrics:**
 - Circles could be added (though not explicitly in the code snippet) to indicate regions around the launch sites, such as safety zones or areas of interest.
- **Lines Connecting Launch Sites to Nearest Railways:**
 - Polyline objects (green lines) were drawn between each launch site and the nearest railway coordinates provided in the DataFrame.
- **Distance Annotations:**
 - Although not explicitly shown on the map, distances between the launch sites and nearby infrastructure (e.g., railways, coastlines) were calculated and could be added as tooltip annotations.
- **Highlight of Key Infrastructure:**
 - By plotting both the launch sites and their closest infrastructure (like railways), the map emphasizes the relationship and accessibility of these points.
- Add the GitHub URL of your completed interactive map with Folium map, as an external reference and peer-review purpose

Build a Dashboard with Plotly Dash

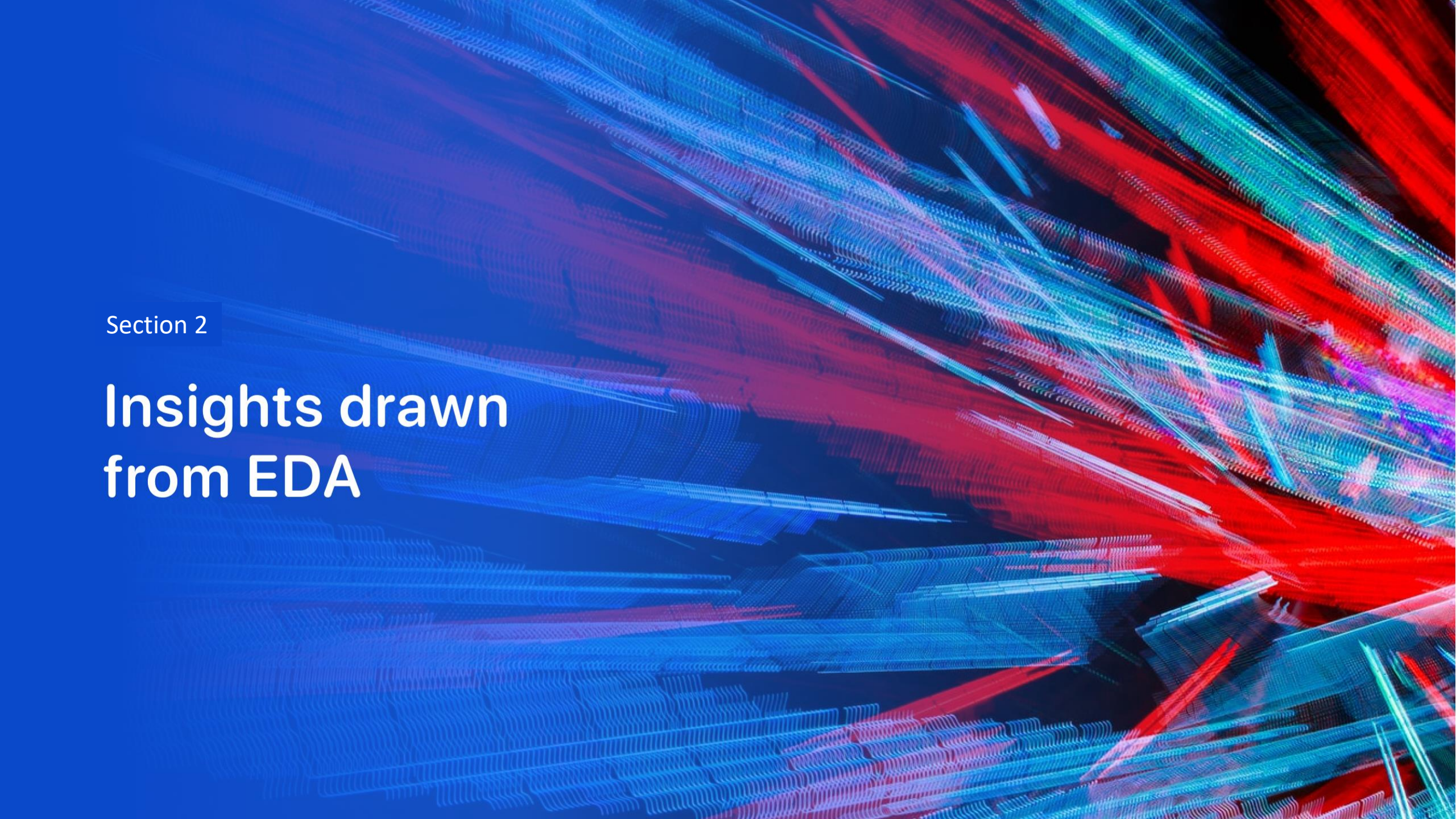
- **Dropdown for Launch Site Selection:**
 - A dropdown menu allows users to filter data by selecting a specific launch site or viewing all sites.
- **Success Pie Chart:**
 - A pie chart updates dynamically based on the selected launch site.
 - **For all sites:** Displays the proportion of successful launches across all launch sites.
 - **For a specific site:** Shows the distribution of successful vs. failed launches for the selected site.
- **Payload Range Slider:**
 - A range slider allows users to filter the data based on payload mass, with a minimum of 0 kg and a maximum of 10,000 kg, adjustable in 1,000 kg increments.
- **Success-Payload Scatter Plot:**
 - A scatter plot visualizes the relationship between payload mass and launch success (`class`), with points colored by booster version.
 - The plot updates dynamically based on the selected launch site and the payload range.
- **Dynamic Interactions:**
 - The dashboard updates both the pie chart and scatter plot in response to user inputs, such as dropdown selections or slider adjustments.
- Add the GitHub URL of your completed Plotly Dash lab, as an external reference and peer-review purpose

Predictive Analysis (Classification)

- Looked at various different models to predict the success of the booster landing.
- Fine tuned hyperparameters using GridSearchCV
- Displayed results using a confusion matrix and presented the accuracy
- **Logistic Regression**
 - **Best parameters:** {'C': 0.01, 'penalty': 'l2', 'solver': 'lbfgs'}
 - **Accuracy:** 0.846
- **Support Vector Machine**
 - **Best parameters:** {'C': 1.0, 'gamma': 0.03162277660168379, 'kernel': 'sigmoid'}
 - **Accuracy:** 0.8482
- **Decision Tree**
 - **Best parameters:** {'criterion': 'entropy', 'max_depth': 14, 'max_features': 'sqrt', 'min_samples_leaf': 2, 'min_samples_split': 5, 'splitter': 'best'}
 - **Accuracy:** 0.9444
- Add the GitHub URL of your completed predictive analysis lab, as an external reference and peer-review purpose

Results

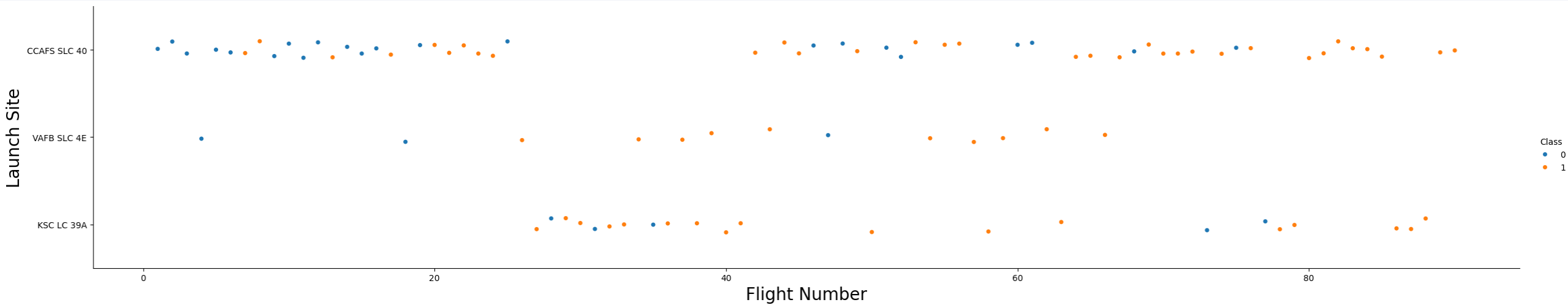
- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower half of the image. The overall effect is dynamic and technological.

Section 2

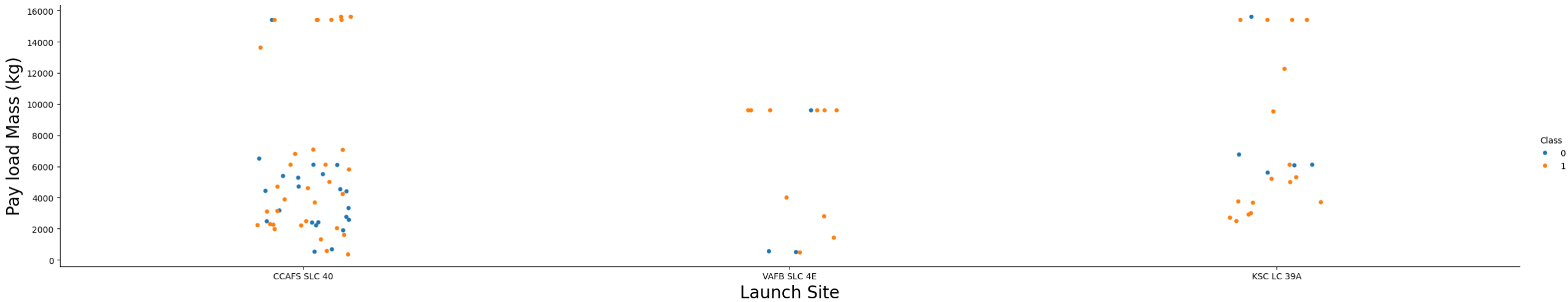
Insights drawn from EDA

Flight Number vs. Launch Site



- While there were quite a few more launches from CCAFS SLC 40, there appears to be about the same percentage of successes no matter the launch site

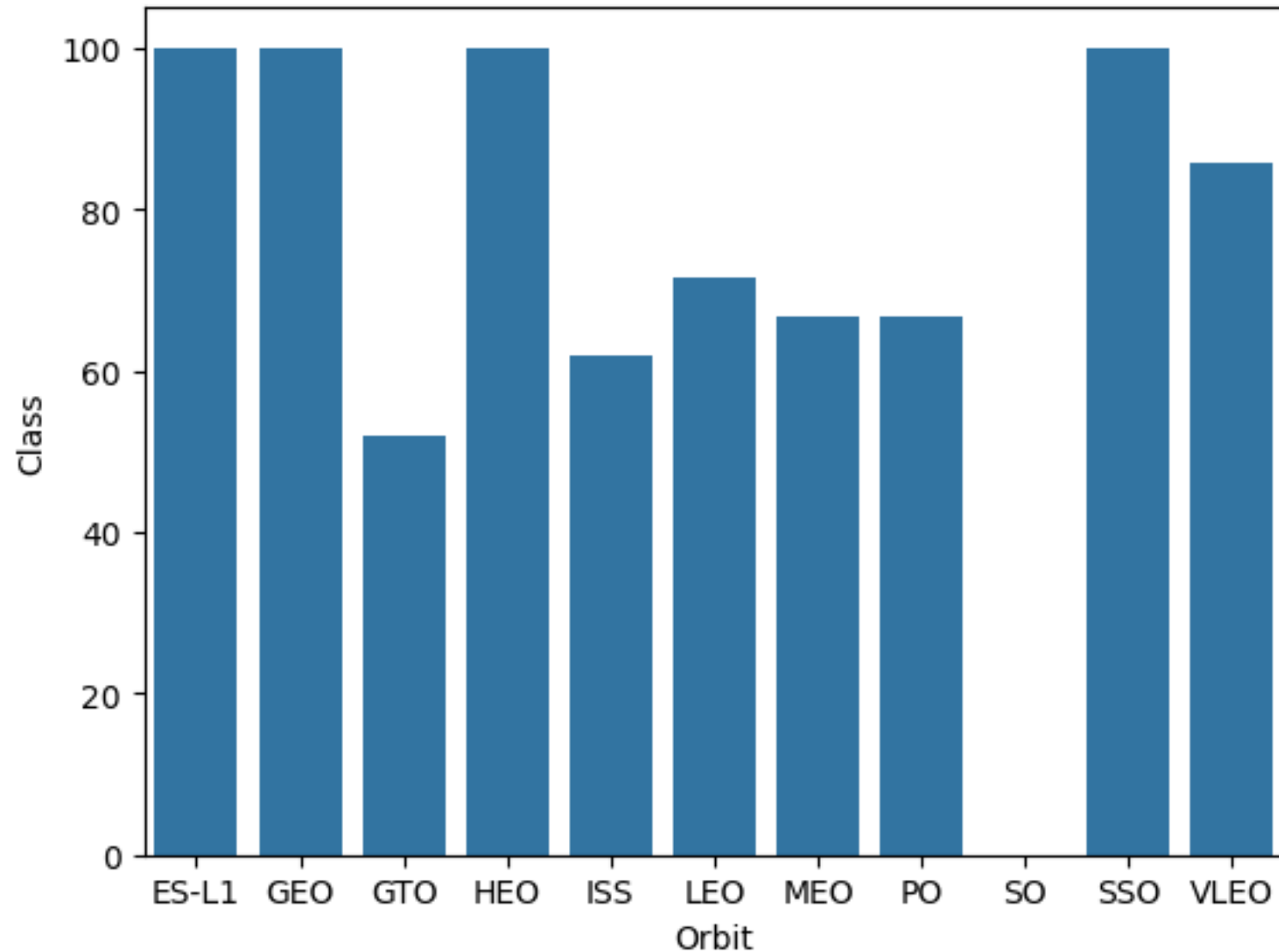
Payload vs. Launch Site



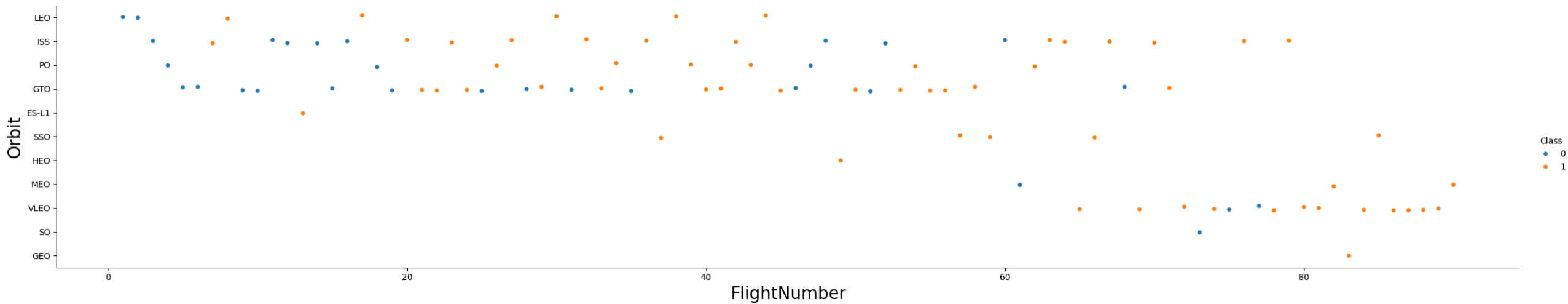
- While there are fewer launches for payloads greater than 9,000 KG, there appears to be a better success rate for these launches.

Success Rate vs. Orbit Type

- The Graph shows that the orbits of ES-L1, GEO, HEO, and SSO have significantly better success rates than the other orbit types.

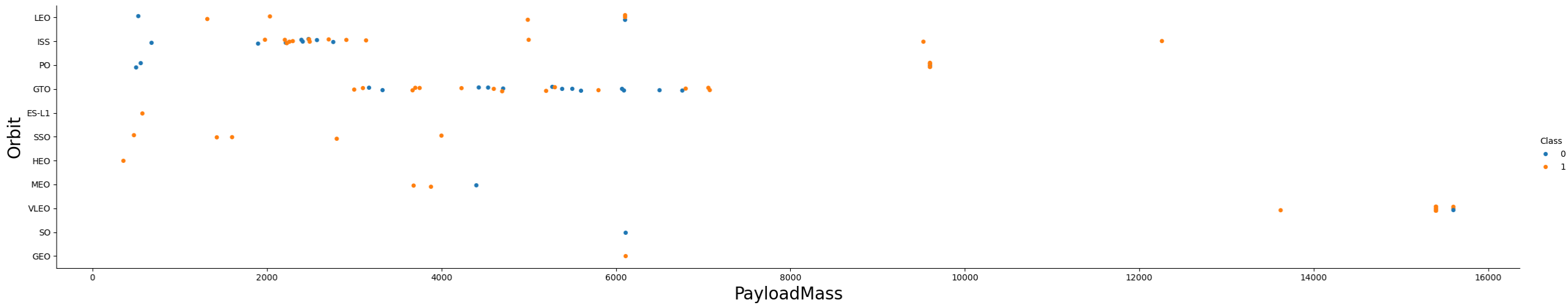


Flight Number vs. Orbit Type



- The scatter plot shows that the success rate of a launch got better as time went on, suggesting that they learned from their previous launches.

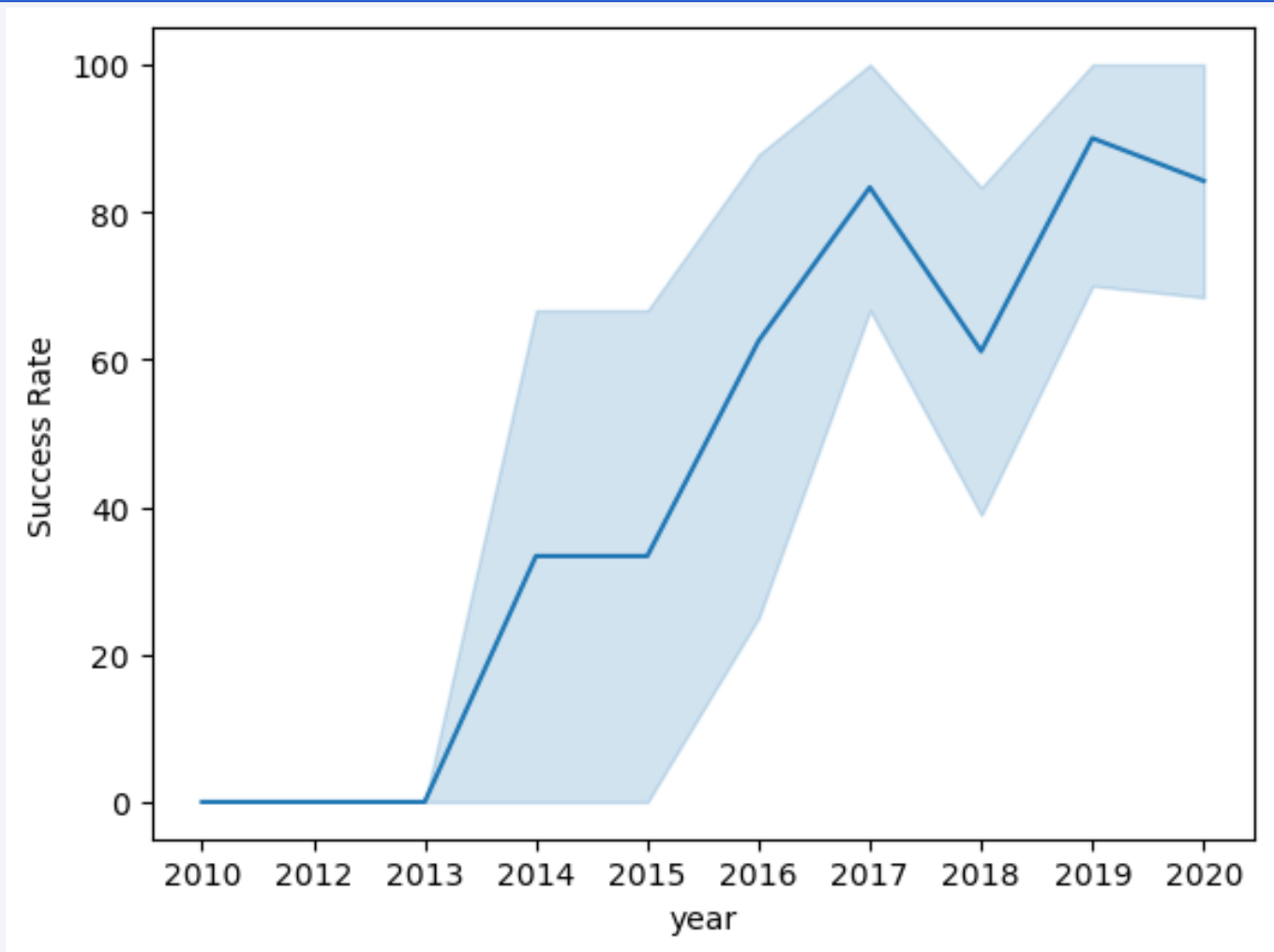
Payload vs. Orbit Type



- The scatter plot shows that most of the higher payloads ($> 9,000$ KG) where in the ISS, PO, and VLEO orbits.

Launch Success Yearly Trend

- The line chart clearly shows that successful launches increased as time went on, suggesting that they learned from previous launches.



All Launch Site Names

- The following SQL statement was used to find the unique launch sites

```
SELECT DISTINCT LAUNCH_SITE from SPACEXTABLE
```

- This query uses the SPACEXTABLE to find the distinct names within the LAUNCH_SITE column.
- This produces the following results:

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

- The following SQL query finds 5 records where launch sites begin with 'CCA'

```
SELECT * from SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5
```

- This query uses the SPACEXTABLE to find the rows where the Launch_Site begins with 'CCA' and only returns the first 5 rows.
- The results are shown below:

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- The following SQL statement was used to calculate the total payload carried by boosters from NASA

```
SELECT SUM(PAYLOAD_MASS__KG_) AS Sum from SPACEXTABLE WHERE Customer like 'NASA (CRS)'
```

- This query uses the SPACEXTABLE to sum all of the values in the PAYLOAD_MASS_KG_ column where the launches were for NASA
- Below are the results of this query:

Sum
45596

Average Payload Mass by F9 v1.1

- The following query was used to calculate the average payload mass carried by booster version F9 v1.1

```
SELECT AVG(PAYLOAD_MASS__KG_) AS Avg from SPACEXTABLE WHERE Booster_Version like 'F9 v1.1%'
```

- The query uses the SPACEXTABLE to average the values of the PAYLOAD_MASS_KG_ column where the Booster_Version is the F9 v1.1
- The results are shown below:

Avg

2534.6666666666665

First Successful Ground Landing Date

- The following query was used to find the dates of the first successful landing outcome on ground pad

```
SELECT min(date) AS Date FROM SPACEXTABLE WHERE mission_outcome LIKE 'Success'
```

- This query uses the SPACEXTABLE to find the earliest date where the mission_outcome was 'Success'
- The results are shown below:

Date

2010-06-04

Successful Drone Ship Landing with Payload between 4000 and 6000

- The following query finds the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

```
SELECT Booster_Version  
FROM SPACEXTABLE  
WHERE (Mission_Outcome LIKE '%Success%') AND  
      (PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000) AND  
      (Landing_Outcome LIKE '%Success (drone ship)%')
```

- This query uses the SPACEXTABLE to find where the Mission_Outcome was a success and the Landing_Outcome was a Success (drone ship) and the payload was between 4000Kg and 6000Kg
- The results are below

```
Booster_Version  
F9 FT B1022  
F9 FT B1026  
F9 FT B1021.2  
F9 FT B1031.2
```


Total Number of Successful and Failure Mission Outcomes

- The following query was used to calculate the total number of successful and failure mission outcomes

```
SELECT Mission_Outcome, COUNT(*) AS Number
FROM SPACEXTABLE
GROUP BY Mission_Outcome
ORDER BY Mission_Outcome
```

- This query uses the SPACEXTABLE to count the number of each of the Mission_Outcomes
- The results are below

Mission_Outcome	Number
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

- The following query displays the names of the boosters which carried the maximum payload mass

```
SELECT Booster_Version  
FROM SPACEXTABLE  
WHERE PAYLOAD_MASS__KG_ = (SELECT max(PAYLOAD_MASS__KG_) FROM SPACEXTABLE)
```

- This query uses a sub-query of the SPACEXTABLE to find the maximum value of the PAYLOAD_MASS_KG_ column and uses that answer to find the Booster_Version of that corresponding launches
- The result is shown to the side:

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

- The following query lists the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015

```
SELECT SUBSTR(Date, 6, 2) as Month, Landing_Outcome, Booster_Version, Launch_Site
FROM SPACEXTABLE
WHERE DATE like '2015%' AND
      Landing_Outcome LIKE 'Failure (drone ship)'
```

- The query uses the SPACEXTABLE to find where the Landing_Outcomes are Failure (drone ship) in the year 2015
- The output is shown below:

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- The following query ranks the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

```
SELECT Landing_Outcome, COUNT(*) as Number
FROM SPACEXTABLE WHERE Date >= '2010-06-04' AND Date <= '2017-03-20'
GROUP BY Landing_Outcome
ORDER BY Number Desc
```

- The query uses the SPACEXTABLE to find the Landing_Outcome and count of these outcomes for each type of Landing_Outcome and orders them by the number of Landing_Outcomes in descending order.

- The results of this query are shown below:

Landing_Outcome	Number
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

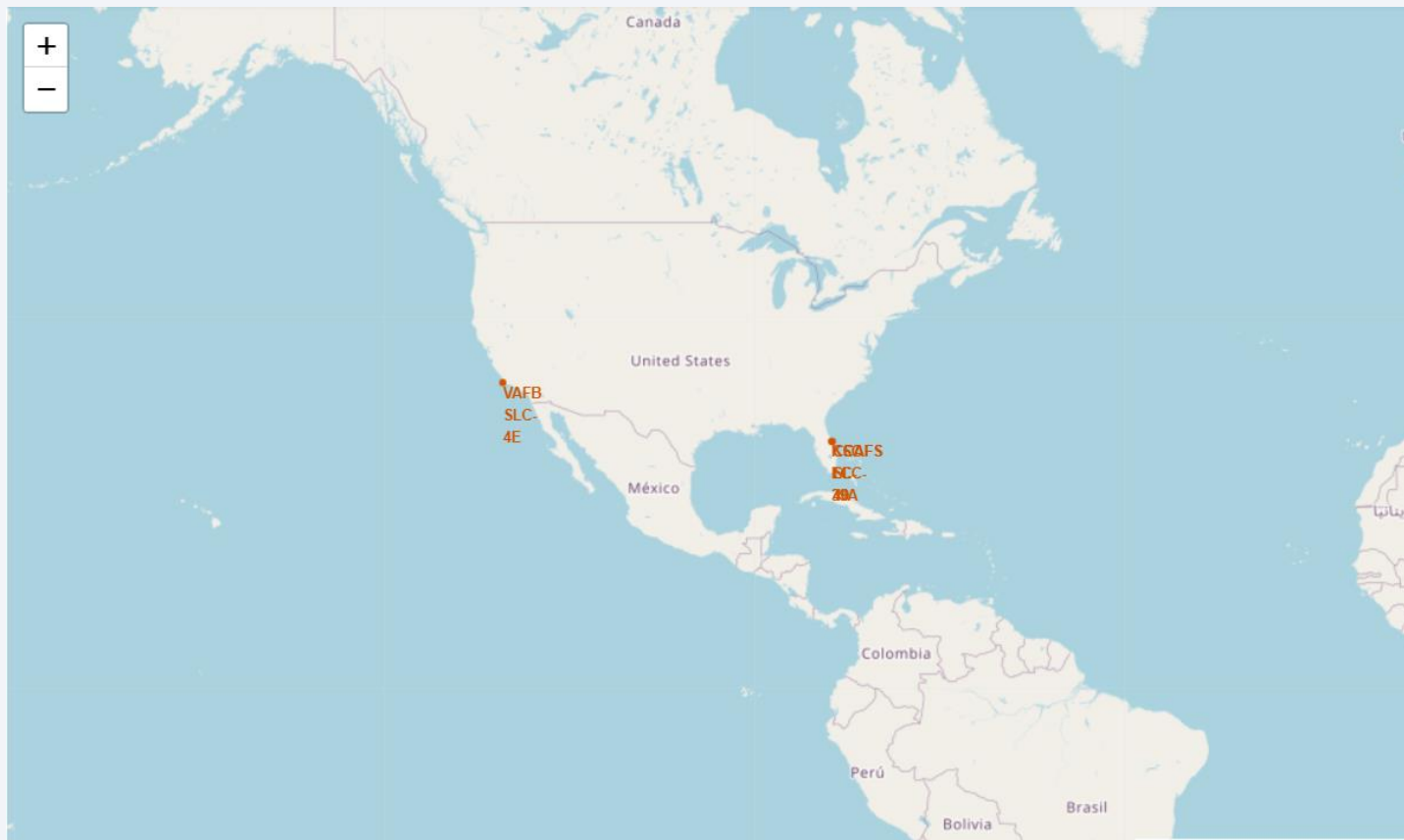
A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

Launch Sites Proximities Analysis

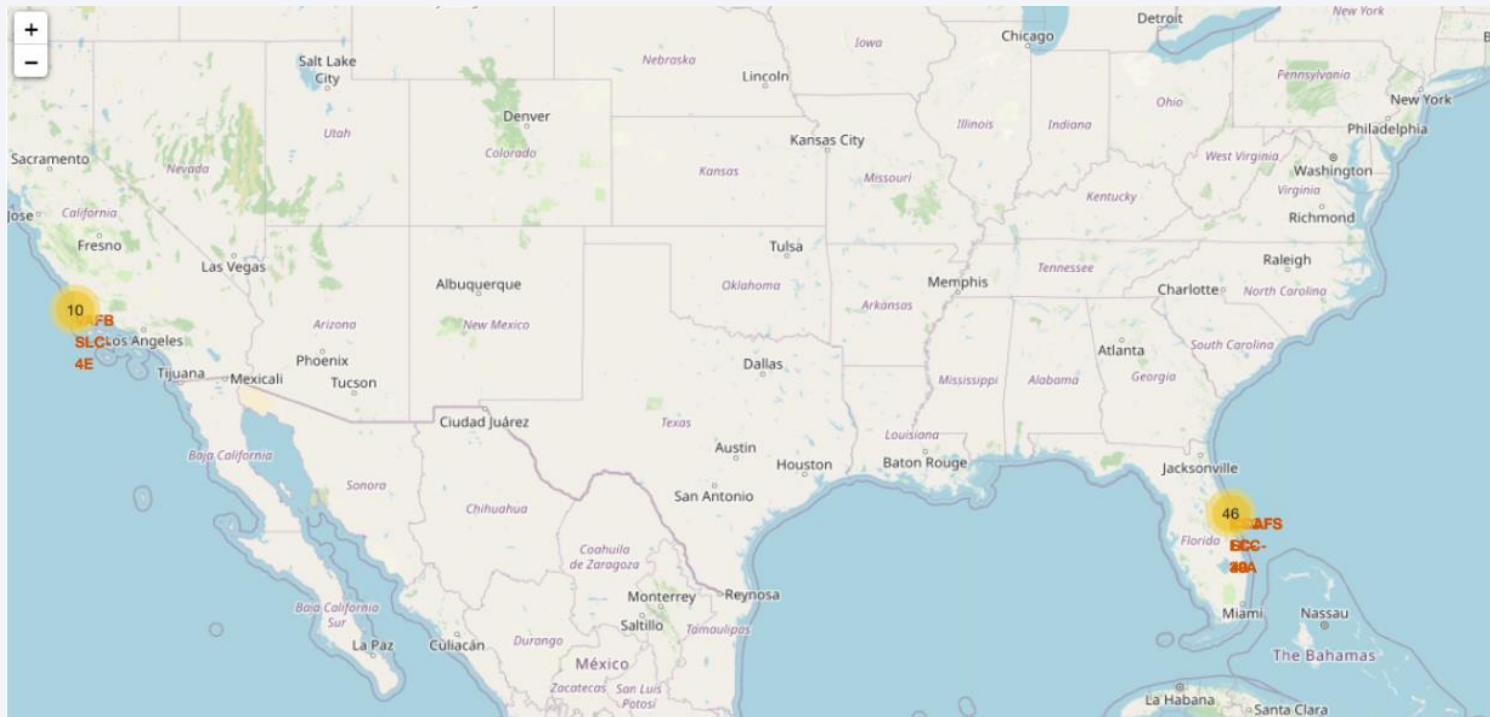
Map of Launch Sites

- As can be seen in the map below, all of the launch sites are in close proximity to the ocean. Three launch sites are located in Florida, while one is located in California.



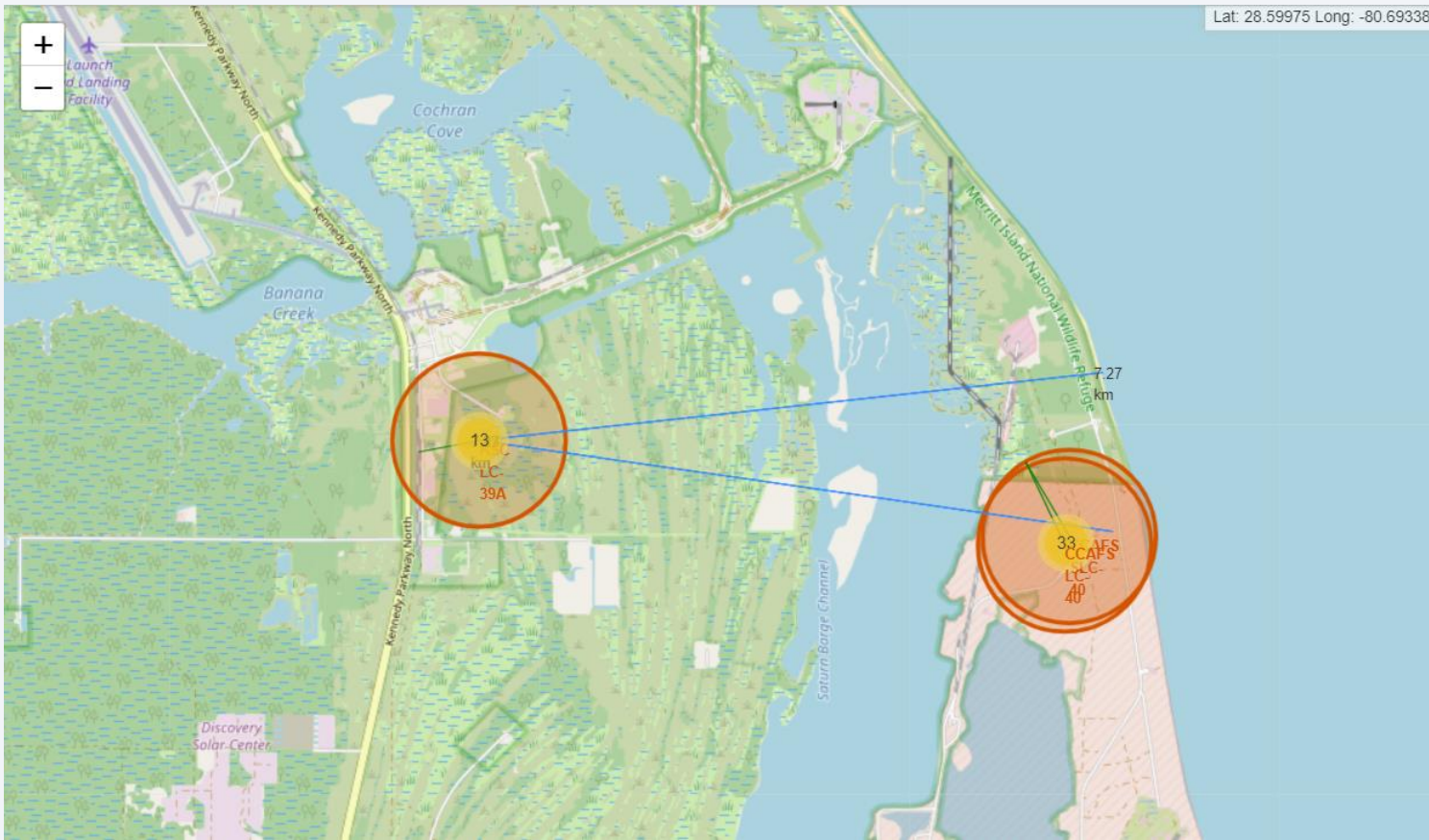
Map of launches

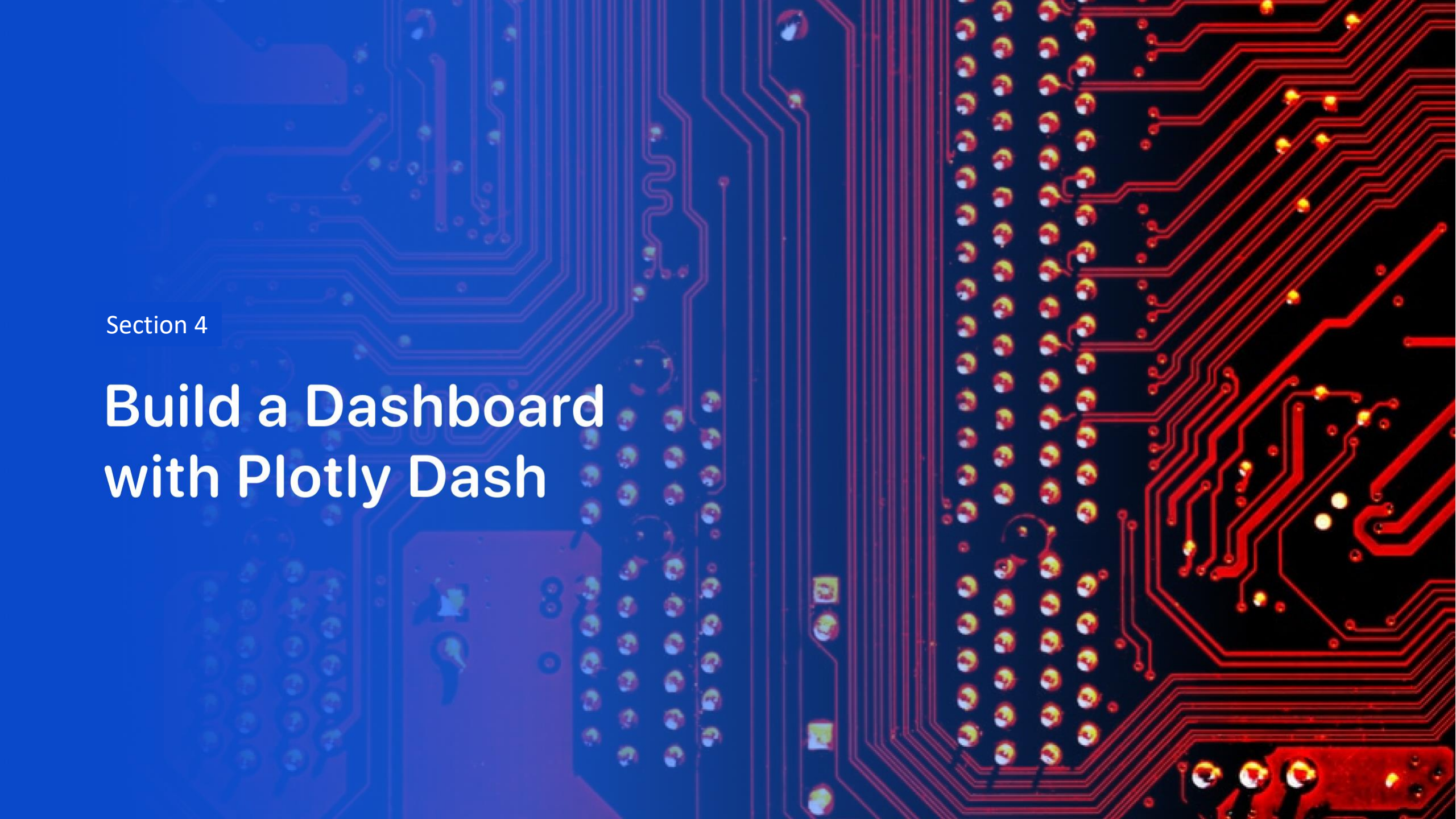
- As can be seen in the below map, most of the launches (46) occurred in Florida, While 10 occurred in California.



Distance to Coastline

- Folium has the capability to show distances to different landmarks. Here we can see launch site is 7.27KM from the coastline



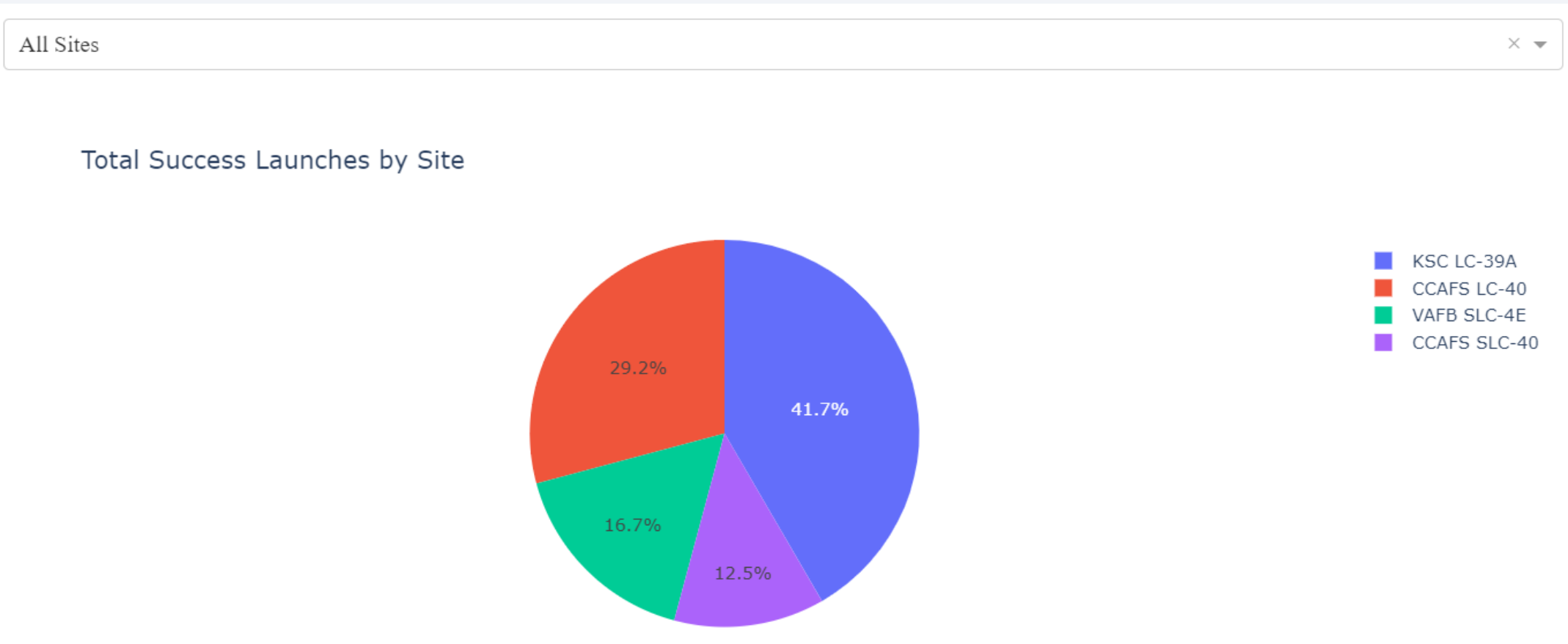


Section 4

Build a Dashboard with Plotly Dash

Launch Success Pie Chart

- The Pie chart shows that the launch site KSC LC-39A has the highest success rate, while launch site CCAFS SLC-40 has the lowest success rate.



Plot of Success of Different Payload and Booster

- The below chart shows that most of the successful missions used the F1 Booster

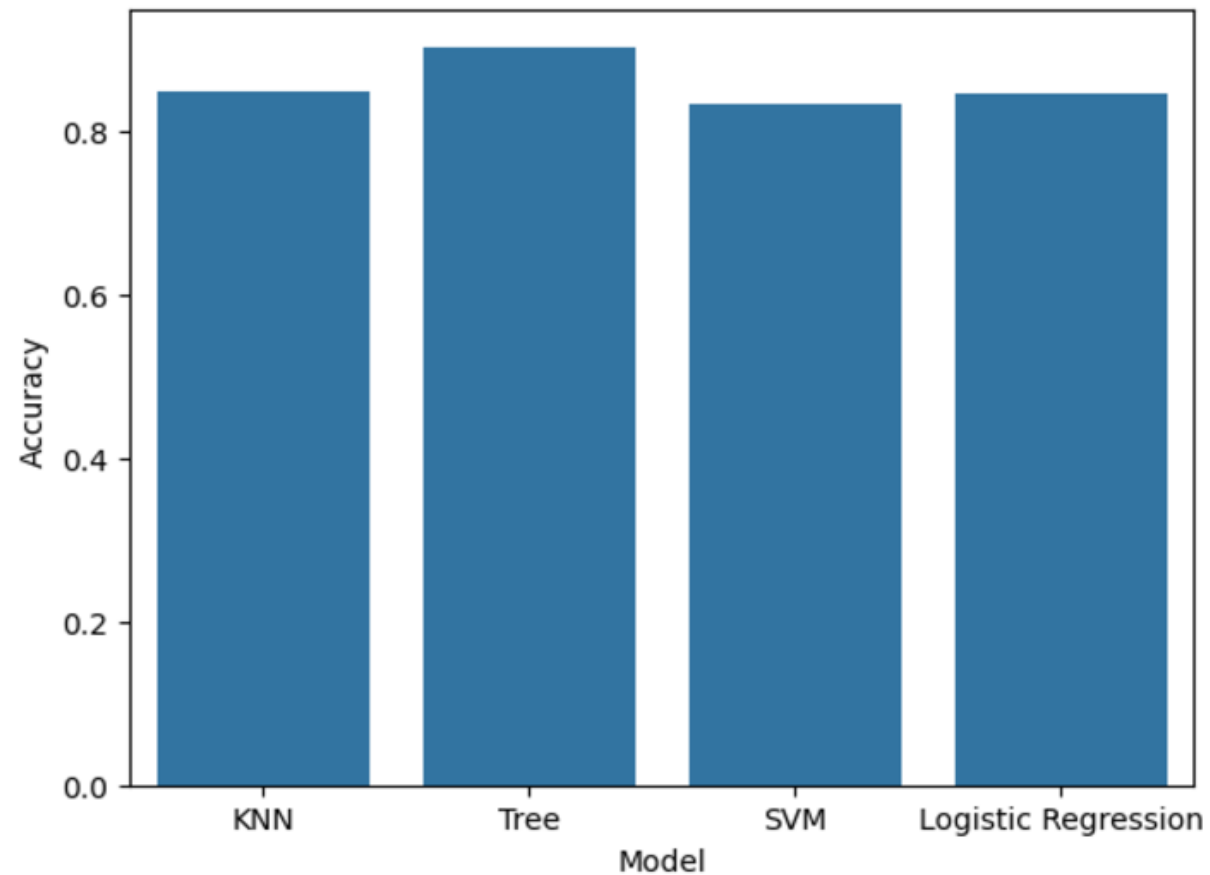


Section 5

Predictive Analysis (Classification)

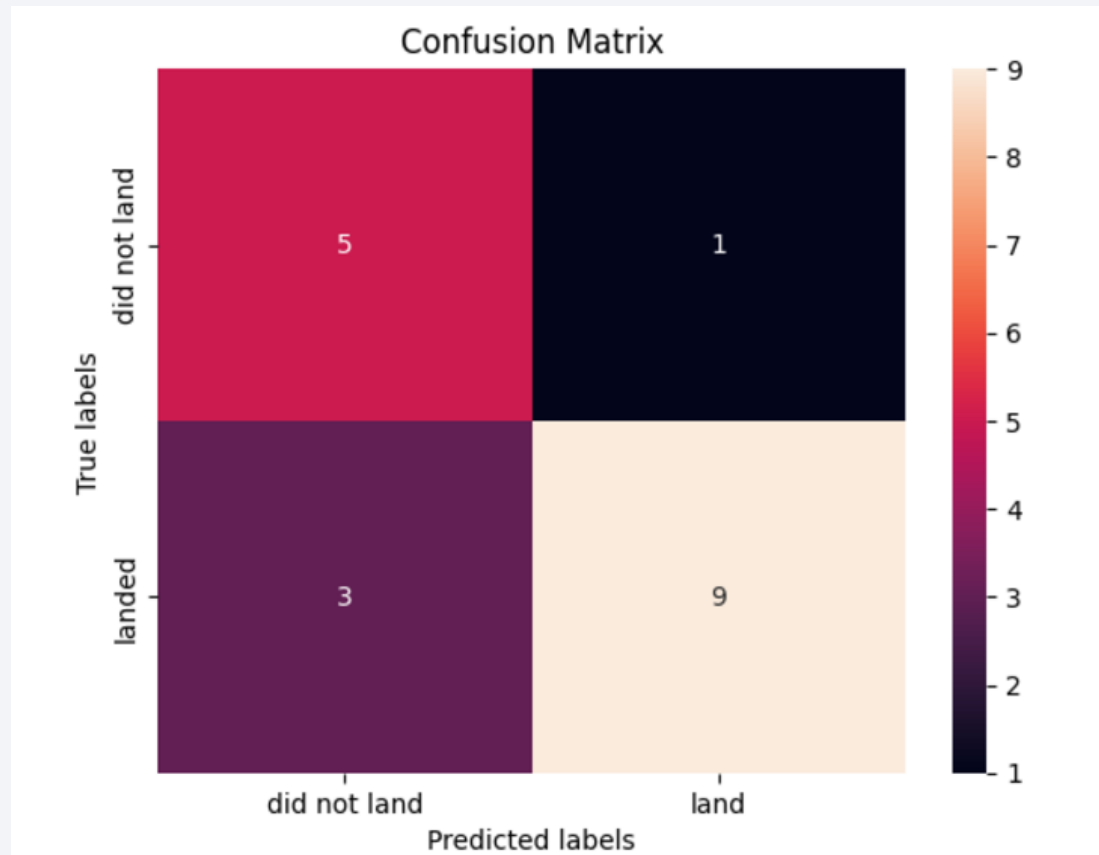
Classification Accuracy

- For classifying the success of a mission landing, the CV_Tree model performed the best with almost an 90% accuracy.



Confusion Matrix

- The confusion matrix of the decision tree classifier shows that the classifier can distinguish between different categories, however the model sometimes struggles to correctly predict a successful landing (False Negatives)



Conclusions

- As the number of flights at a launch site increase, the success rate at that site increases, suggesting that there is a learning curve in order to successfully land the booster.
- The overall launch success also can be seen to increase as the number of overall flights increases, again suggesting that the teams learned through their past launches.
- The ES-L1, GEO, HEO, SSO, and VLEO orbits have the highest success rates.
- The decision tree model was the best predictor of landing success, though other models performed similarly.
- Insights from this analysis can help estimate launch costs and improve future predictions. With a larger dataset and further feature analysis, these models could become even more accurate, offering valuable information for competitive bidding in the space industry.

Thank you!

